Sales Pulse – Sales and Demand Forecasting System

A Project Report Submitted in

the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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AY- 2026-27

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Declaration

The Project Report entitled "**SALES PULSE**" is a record of Bonafide work of **Y. Nimisha – 2420030427, K. Sanuhya – 2420030498, R. Sarvajith Varma – 2420030516** submitted in partial fulfillment for the award of B. Tech in CSE to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the project report entitled "SALES PULSE – Sales and Demand Forecasting System" is a bonafide work carried out and submitted by Y. Nimisha (2420030427), K. Sanuhya (2420030498), R. Sarvajith varma (2420030516) in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of Computer Science and Engineering, K L (Deemed to be) University, during the academic year 2026–2027.

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ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my Course Coordinator and my guide **Dr. N. Sunanda,** Department of CSE which cannot be expressed in words for her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P. Venkateswara Rao**, Head of the ***Department for*** **COMPUTER SCIENCE AND ENGINEERING** for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

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Abstract

Businesses generate large amounts of historical sales data through daily transactions. Analyzing this data helps understand sales trends, customer purchasing patterns, demand variations, and seasonal changes. However, traditional sales management systems mainly focus on recording past transactions and generating reports, with limited capabilities to predict future sales. This makes it difficult for businesses to identify high- and low-demand periods and plan inventory effectively.

This paper presents Sales Pulse, a sales and demand forecasting system designed to convert historical sales data into future predictions and meaningful business insights. The system involves collecting and preprocessing sales datasets, performing Exploratory Data Analysis (EDA) to identify trends and seasonal patterns, and applying time-series forecasting techniques such as ARIMA and Prophet. Forecasting performance is evaluated using accuracy metrics such as MAE, MSE, and RMSE to measure prediction reliability.

Sales Pulse provides an interactive dashboard to display historical sales, predicted values, graphical visualizations, and important business insights. Users can analyze sales across different time periods and understand changes in product demand. The system aims to help businesses predict future sales, improve inventory planning, reduce overstocking and stock shortages, and make better data-driven decisions. Overall, Sales Pulse supports proactive business planning and improves sales management efficiency.

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CHAPTER 1

Introduction

1.1 Motivation

The main motivation behind Sales Pulse is to help businesses understand their sales performance and predict future demand using historical sales data. Traditional sales management systems mainly focus on recording past transactions and generating reports, with limited forecasting capabilities. Sales Pulse aims to overcome this limitation by analyzing historical sales patterns, identifying seasonal variations, and predicting future sales. This helps businesses understand high- and low-demand periods and make better decisions regarding inventory and sales planning.

1.2 Objective

The main objective of Sales Pulse is to develop a sales and demand forecasting system that converts historical sales data into meaningful insights and future predictions. The system uses data preprocessing, Exploratory Data Analysis (EDA), and time-series forecasting techniques such as ARIMA and Prophet. It also provides an interactive dashboard to display historical sales, forecasted values, and visual insights. This helps businesses improve inventory management, reduce stock shortages and overstocking, and make informed data-driven decisions.

1.3 What is Data Science?

Data Science is an interdisciplinary field that involves collecting, processing, analyzing, and interpreting data to extract meaningful information. It combines statistics, programming, and machine learning techniques to identify patterns, discover trends, and make predictions. Data Science plays an important role in business decision-making by converting raw data into useful insights.

Data Science involves various activities such as data collection, data preprocessing, data analysis, data visualization, and predictive modeling. It uses programming languages, statistical methods, and machine learning algorithms to extract valuable information from large datasets.

Data Science can be applied in different fields, including business, healthcare, finance, education, and retail. In Sales Pulse, these techniques are used to process sales datasets, identify patterns, analyze demand variations, and generate future sales predictions.

1.4 What is Sales Forecasting?

Sales Forecasting is the process of estimating future sales based on historical sales data, trends, and patterns. It helps businesses understand expected demand and prepare for future requirements. Forecasting techniques such as ARIMA and Prophet analyze past observations to predict future sales values.

Sales forecasting plays an important role in business planning because it helps organizations estimate product requirements, manage inventory, and identify potential changes in demand. In Sales Pulse, sales forecasting is used to identify high- and low-demand periods, support inventory planning, and improve business decision-making.

1.5 What is a Dashboard?

A Dashboard is a visual interface that presents important information and data through charts, graphs, tables, and other visual elements. It helps users understand complex information quickly and monitor performance. Dashboards allow users to compare data, identify trends, and gain useful insights.

A dashboard provides a simple and organized way to access important information without manually analyzing large amounts of data. In Sales Pulse, the dashboard displays historical sales data, forecasted sales values, graphical visualizations, and important business insights. It helps users understand sales performance, monitor demand variations, and make informed decisions based on the available data.

1.6 What is Time-Series Analysis?

Time-Series Analysis is a statistical technique used to analyze data collected over a specific period. It helps identify patterns such as trends, seasonal variations, and fluctuations in data over time. Time-series analysis is widely used in sales forecasting, weather prediction, finance, and business planning.

In Sales Pulse, time-series analysis is applied to historical sales records to understand how sales change over time. Forecasting models such as ARIMA and Prophet use these historical patterns to estimate future sales and demand. This helps businesses prepare for upcoming demand changes and improve their planning strategies.

CHAPTER 2

Software Requirements

2.1 Problem Statement

Businesses generate large amounts of sales data, but traditional systems mainly store historical records without predicting future demand. This makes it difficult to identify sales trends, seasonal variations, and high- or low-demand periods. Sales Pulse addresses this problem by analyzing historical sales data using forecasting techniques such as ARIMA and Prophet.

2.2Technologies used

1. Python:

Python is a high-level, interpreted programming language used for data analysis, machine learning, and data visualization. It provides a wide range of libraries that simplify data processing and forecasting. In the Sales Pulse project, Python is used to clean sales datasets, perform exploratory data analysis, build forecasting models, and generate sales predictions.

General Syntax of Python:

Simple Python Programprint("Welcome to Sales Pulse")sales = [1000, 1500, 2000, 2500, 3000]total_sales =

sum(sales)average_sales = total_sales / len(sales)print("Total Sales:", total_sales)print("Average Sales:", average_sales)

Output:

Welcome to Sales Pulse

1. Total Sales: 10000
2. Average Sales: 2000.0

2. Pandas:

Pandas is a Python library used for data manipulation and analysis. It provides data structures such as Series and DataFrames to organize and process structured data. In Sales Pulse, Pandas is used to load sales datasets, handle missing values, remove duplicate records, and prepare data for forecasting.

General Syntax of Pandas:

import pandas as pddata = { 'Date': ['2026-01-01', '2026-01-02', '2026-01-03'], 'Sales': [1000, 1500, 2000]}df = pd.DataFrame(data)print(df)print("Total Sales:", df['Sales'].sum())

Output:

Date Sales

1. 2026-01-01 1000
2. 2026-01-02 1500
3. 2026-01-03 2000

Total Sales: 4500

3. NumPy:

NumPy is a Python library used for numerical computations and mathematical operations. It supports arrays, statistical calculations, and efficient processing of numerical data. In Sales Pulse, NumPy helps perform numerical operations on sales values during data analysis and forecasting.

General Syntax of NumPy:

import numpy as npsales = np.array([1000, 1500, 2000, 2500, 3000])print("Sales Values:", sales)print("Average Sales:", np.mean(sales))print("Maximum Sales:", np.max(sales))

Output:

Sales Values: [1000 1500 2000 2500 3000]

Average Sales: 2000.0

Maximum Sales: 3000

4. Matplotlib:

Matplotlib is a Python library used for creating graphs and visualizations. It helps represent numerical data through line charts, bar charts, and other graphical formats. In Sales Pulse, Matplotlib is used to visualize historical sales trends and forecasted sales values.

General Syntax of Matplotlib:

```
import matplotlib.pyplot as pltmonths = ['Jan', 'Feb', 'Mar', 'Apr', 'May']sales = [1000, 1500, 1200, 2000, 2500]plt.plot(months, sales, marker='o')plt.title("Monthly Sales Analysis")plt.xlabel("Months")plt.ylabel("Sales")plt.show()
```

5. ARIMA:

ARIMA (AutoRegressive Integrated Moving Average) is a statistical time-series forecasting model used to predict future values based on historical data. It uses past observations, differencing, and moving averages to identify patterns. In Sales Pulse, ARIMA is considered for forecasting future sales based on historical sales records.

6. Prophet:

Prophet is a time-series forecasting tool developed by Meta. It is designed to handle trends, seasonality, and changes in time-series data. In Sales Pulse, Prophet is considered as another forecasting approach to predict future sales and identify demand patterns.

7. Streamlit:

Streamlit is an open-source Python framework used to create interactive web applications for data science and machine learning projects. In Sales Pulse, Streamlit is used to develop the dashboard where users can upload sales datasets, select forecasting periods, view charts, and understand sales predictions and business insights.

CHAPTER 3

Software Design

3.1 DFD

The following DFD represents the overall working of the Sales Pulse system. The user uploads sales datasets, and the system processes the data, performs sales analysis and forecasting, and displays the results through an interactive dashboard.

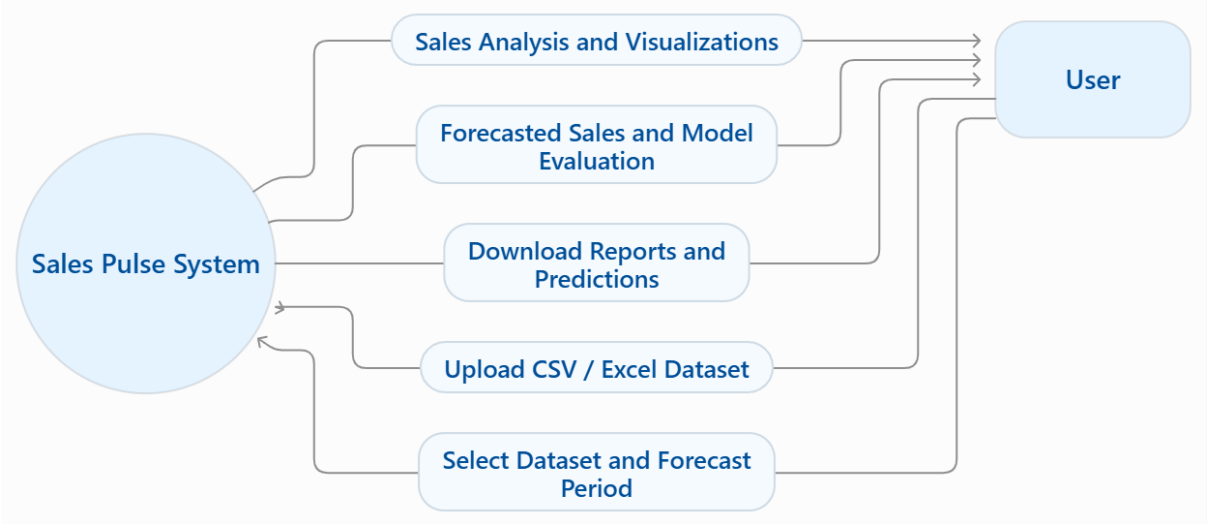


Fig No. 3.1 DFD LEVEL-0

CHAPTER 4

Implementation and User Interface

4.1 Web Page Details

1. Home Page
2. About Sales Pulse
3. Upload Sales Dataset
4. Sales Analysis and Forecasting
5. Dashboard and Business Insights

4.2 Detailed Description of Pages

1. Home Page The Home Page is the main screen of Sales Pulse. It provides an overview of the system, its features, and key sales information. Users can navigate to data upload, sales analysis, forecasting, and dashboard sections.

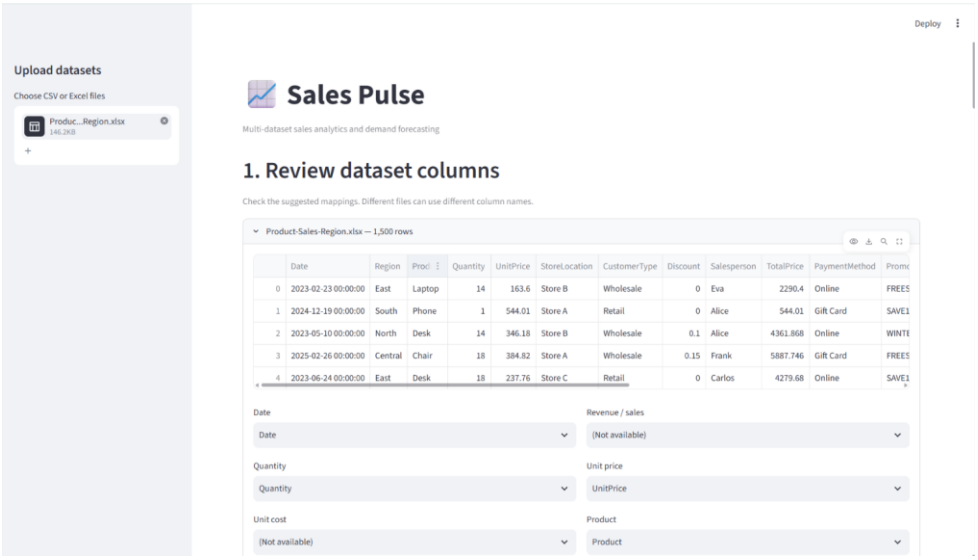


Fig No. 4.1 Home Page

2. Upload Dataset Page : This page allows users to upload sales datasets in CSV or Excel format. The uploaded data is validated and preprocessed before being used for sales analysis and forecasting.

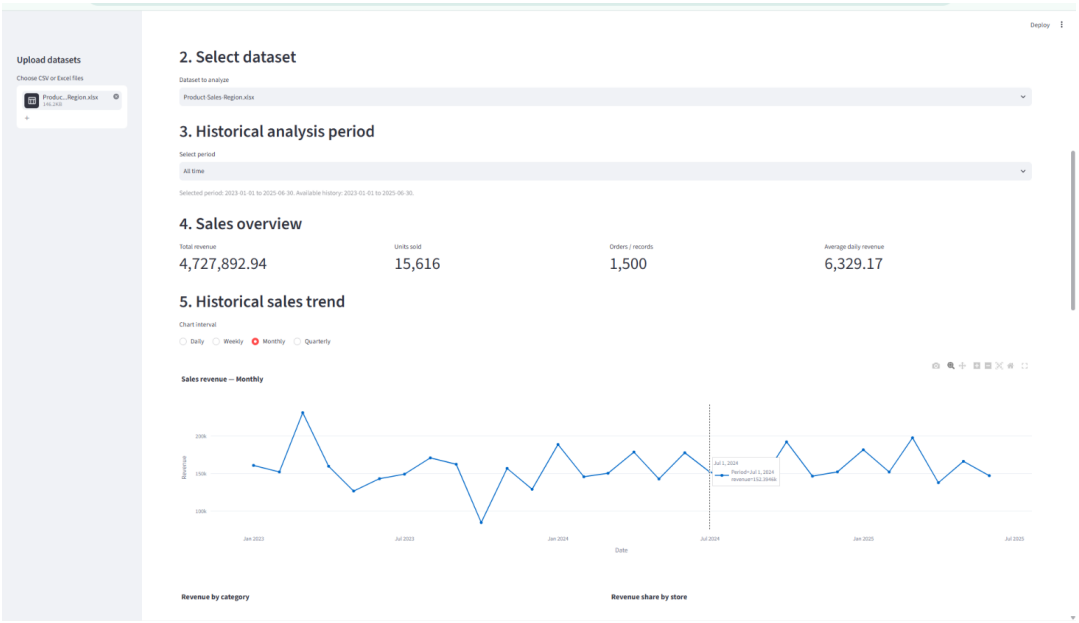


Fig No. 4.3 2. Top 6 Restaurant Page

3. Sales Analysis and Forecasting Page : This page displays historical sales trends and allows users to select a forecasting period. Forecasting models such as ARIMA and Prophet are used to predict future sales based on the uploaded dataset.

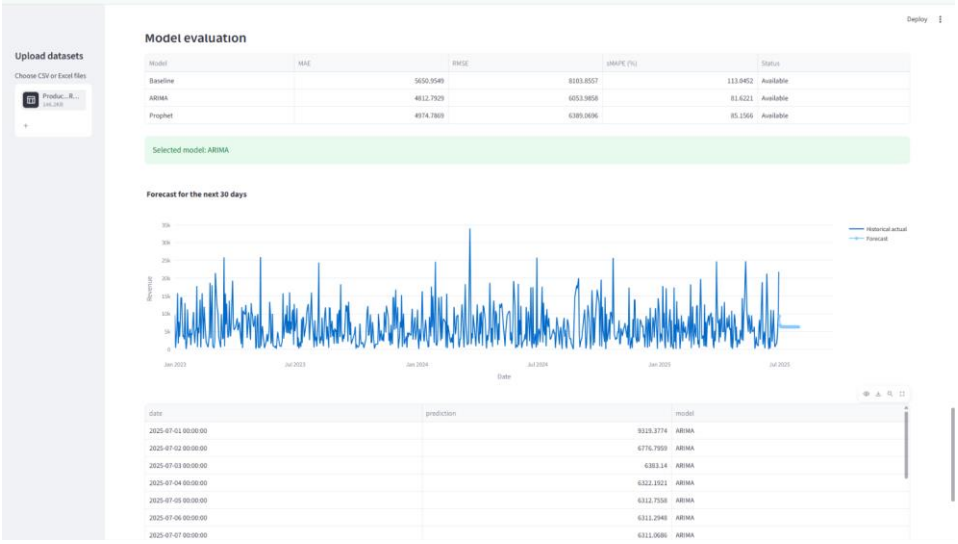


Fig 4.3 Products Page

4. Dashboard and Insights Page : This page displays sales performance through interactive charts, graphs, and key business insights. Users can understand sales patterns, identify high- and low-demand periods, and view forecast results to support better business decisions.



Fig 4.5 Review of the Day Page

CHAPTER 5

Conclusion and Future Aspects

5.1 Conclusion

We have successfully developed the Sales Pulse project, a sales analysis and demand forecasting system. It uses Python libraries and forecasting techniques such as ARIMA and Prophet to analyze historical sales data and predict future demand. The interactive dashboard presents sales trends, forecasted values, and business insights. Overall, the system helps businesses improve inventory management, understand sales patterns, and make better data-driven decisions.

5.2 Future Aspects

1. We can integrate AI-based forecasting models to improve prediction capabilities.
2. We can add real-time sales data integration to provide updated sales insights.
3. We can include automated alerts for high- and low-demand periods.
4. We can add downloadable reports in PDF and Excel formats.
5. We can improve the dashboard with advanced visualizations and product-wise sales analysis.

CHAPTER 6

Bibliography

- https://www.python.org/doc/
- https://pandas.pydata.org/docs/
- https://numpy.org/doc/
- https://matplotlib.org/stable/
- https://www.statsmodels.org/stable/
- https://facebook.github.io/prophet/docs/
- https://docs.streamlit.io/

CHAPTER 7

Appendices

The uploaded content contains HTML, CSS, Bootstrap, and restaurant-review code. Since your project is Sales Pulse, the corresponding implementation should use Python, Streamlit, Pandas, Matplotlib, and forecasting libraries such as ARIMA and Prophet.

Below is the adapted source-code section with the same report-style numbering.

7.1 Python – Required Libraries

The following libraries are used for data preprocessing, sales analysis, visualization, and forecasting.

```
import streamlit as st

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from statsmodels.tsa.arima.model import ARIMA

from sklearn.metrics import mean_absolute_error, mean_squared_error
```

7.2 Page Configuration

This code defines the title, layout, and appearance of the Sales Pulse dashboard.

```
st.set_page_config(

  page_title="Sales Pulse",
```

```
    page_icon="📊",
    layout="wide"
)
st.title("Sales Pulse")
st.subheader("Sales Analysis and Demand Forecasting")
st.write(
    "Analyze historical sales data and predict future demand."
)
```

7.3 Navigation Bar

The sidebar allows users to navigate between different modules of the application.

```
st.sidebar.title("Sales Pulse")
page = st.sidebar.radio(
    "Navigation",
    [
        "Home",
        "Upload Dataset",
        "Sales Analysis",
        "Forecasting",
```

```
        "Dashboard"
    ]
)
```

7.4 Upload Dataset

This section allows users to upload sales datasets in CSV or Excel format.

```
st.header("Upload Sales Dataset")
uploaded_file = st.file_uploader(
    "Choose a CSV or Excel file",
    type=["csv", "xlsx"]
)
if uploaded_file is not None:
    if uploaded_file.name.endswith(".csv"):
        df = pd.read_csv(uploaded_file)
    else:
        df = pd.read_excel(uploaded_file)
    st.success("Dataset uploaded successfully!")
    st.write("Dataset Preview")
    st.dataframe(df.head())
```

7.5 Data Preprocessing

This section handles missing values, duplicate records, and date conversion before analysis.

```
if uploaded_file is not None:
    df = df.drop_duplicates()
    df = df.dropna()
    if "Date" in df.columns:
        df["Date"] = pd.to_datetime(
            df["Date"],
            errors="coerce"
        )
        df = df.dropna(subset=["Date"])
    st.subheader("Cleaned Dataset")
    st.dataframe(df.head())
```

7.6 Sales Analysis and Visualization

This section displays sales information using graphs and statistical calculations.

```
if uploaded_file is not None:
    sales_column = st.selectbox(
"Select Sales Column",
```

```
        df.select_dtypes(include=np.number).columns
    )
    total_sales = df[sales_column].sum()
    average_sales = df[sales_column].mean()
    col1, col2 = st.columns(2)
    col1.metric("Total Sales", f"{total_sales:,.2f}")
    col2.metric("Average Sales", f"{average_sales:,.2f}")
    st.subheader("Sales Visualization")
    fig, ax = plt.subplots()
    ax.plot(df[sales_column], marker="o")
    ax.set_title("Sales Trend")
    ax.set_xlabel("Records")
    ax.set_ylabel("Sales")
    st.pyplot(fig)
```

7.7 Sales Forecasting Using ARIMA

ARIMA is used to analyze historical sales values and forecast future sales.

```
if uploaded_file is not None:
    st.header("ARIMA Forecasting")
```

```
    periods = st.number_input(
        "Enter Forecast Periods",
        min_value=1,
        max_value=365,
        value=30
    )
    sales_data = df[sales_column].dropna()
    if len(sales_data) > 10:
        model = ARIMA(
            sales_data,
            order=(1, 1, 1)
        )
        model_fit = model.fit()
        forecast = model_fit.forecast(
            steps=int(periods)
        )
        st.write("Forecasted Sales")
        st.dataframe(
```

```
        pd.DataFrame({
            "Forecast": forecast
        })
    )
```

7.8 Dashboard and Business Insights

The dashboard displays important sales information and helps users understand business performance.

```
if uploaded_file is not None:
  st.header("Sales Dashboard")
  st.metric(
      "Total Sales",
      f"{df[sales_column].sum():,.2f}"
  )
  st.metric(
      "Maximum Sales",
      f"{df[sales_column].max():,.2f}"
  )
  st.metric(
      "Minimum Sales",
```

```
        f"{df[sales_column].min():,.2f}"
    )
    st.subheader("Business Insights")
    st.write(
        "Highest Sales:",
        df[sales_column].max()
    )
    st.write(
        "Lowest Sales:",
        df[sales_column].min()
    )
    st.write(
        "Average Sales:",
        df[sales_column].mean()
    )
```

7.9 Model Evaluation

Forecasting performance can be evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

```
if uploaded_file is not None:
    if len(sales_data) > 10:
        train_size = int(len(sales_data) * 0.8)
        train_data = sales_data[:train_size]
        test_data = sales_data[train_size:]
        if len(test_data) > 0:
            model = ARIMA(
                train_data,
                order=(1, 1, 1)
            )
            fitted_model = model.fit()
            predictions = fitted_model.forecast(
                steps=len(test_data)
            )

            mae = mean_absolute_error(
                test_data,
                predictions
```

```
        )
        rmse = np.sqrt(
            mean_squared_error(
                test_data,
                predictions
            )
        )
        st.write("MAE:", round(mae, 2))
        st.write("RMSE:", round(rmse, 2))
```